

NICHOLAS E. CORRADO

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EDUCATION

- University of Wisconsin – Madison**, Madison, WI Present
Ph.D., Computer Sciences
Dissertation: Data Collection for Sample-Efficient Reinforcement Learning
Advisor: [Josiah Hanna](#)
Keywords: reinforcement learning, adaptive data collection, multi-task learning, robotics, LLM alignment
- University of Pittsburgh**, Pittsburgh, PA 2019
B.S. in Physics, Mathematics; Minor in Computer Science
Thesis: [A Search for \$W_{bJ}\$ in Decays of \$\Upsilon\(5S\)\$: An Analysis Design Study](#)
Advisor: [Vladimir Savinov](#)

EXPERIENCE

- University of Wisconsin–Madison** | Graduate Research Assistant Jan. 2021 – Present
Advisor: Josiah Hanna
- ▶ My research focuses on **adaptive data collection** and **data mixing** for **reinforcement learning**, with applications to **LLM alignment**, **robotics**, and **multi-agent learning**.
- Amazon** | Applied Scientist Intern July 2024 – Nov. 2024
Advisors: Julian Katz-Samuels, Hyokun Yun, Yi Xu
- ▶ **Multi-task alignment for LLMs** with the [Rufus Team](#). I developed a DPO-based algorithm that balances performance across multiple tasks with convergence guarantees. See the [AutoMixAlign paper \(ACL 2025\)](#).
- Sandia National Laboratories** | Graduate Research Intern May 2021 – Nov. 2023
Advisor: Drew Levin
- ▶ Reinforcement learning for **power systems management**.
- University of Wisconsin–Madison** | Graduate Research Assistant Sept. 2019 – Sept. 2020
Advisor: Jignesh Patel
- ▶ Developed efficient **query execution algorithms**. Built the query execution and storage engines for [Hustle](#).

PAPERS

- Manuscripts Under Review
- [M1] **Distributionally Robust Multi-Task Reinforcement Learning via Adaptive Task Sampling**
[Nicholas E. Corrado](#), [Wenyuan Huang](#), [Josiah P. Hanna](#)
Under Review, 2026.
+ 🏆 **Best Paper Award** at Automated RL Workshop @ RLC, 2026. (Oral).
- Conference Publications.....
- [C1] **Centralized Adaptive Sampling for Reliable Co-training of Independent Multi-Agent Policies**
[Nicholas E. Corrado](#), [Josiah P. Hanna](#)
Reinforcement Learning Conference (RLC), 2026.
- [C2] **On-Policy Policy Gradient Reinforcement Learning Without On-Policy Sampling**
[Nicholas E. Corrado](#), [Josiah P. Hanna](#)
Transactions on Machine Learning Research (TMLR), 2026.
+ Journal-to-Conference Track at the Reinforcement Learning Conference (RLC), 2026.
+ Finding the Frame Workshop at the Reinforcement Learning Conference (RLC), 2025.

- [C3] **When Can Model-Free Reinforcement Learning be Enough for Thinking?**
 Josiah P. Hanna, Nicholas E. Corrado
 Neural Information Processing Systems (NeurIPS), 2025.
 + 🏆 **“Most-Thought-Provoking Paper”** at Finding the Frame Workshop, RLC 2025. **(Oral)**.
- [C4] **AutoMixAlign: Adaptive Data Mixing for Multi-Task Preference Optimization in LLMs**
 Nicholas E. Corrado, Julian Katz-Samuels, Adithya M Devraj, Hyokun Yun, Chao Zhang, Yi Xu, Yi Pan, Bing Yin, Trishul Chilimbi
 Association for Computational Linguistics (ACL), 2025. (Acceptance Rate: 20.3%)
 + Amazon Machine Learning Conference (AMLC), 2025 **(Oral)**.
- [C5] **Guided Data Augmentation for Offline Reinforcement Learning and Imitation Learning**
 Nicholas E. Corrado, Yuxiao Qu, John U. Balis, Adam Labiosa, Josiah P. Hanna
 Reinforcement Learning Conference (RLC), 2024.
- [C6] **Understanding when Dynamics-Invariant Data Augmentations Benefit Model-Free Reinforcement Learning Updates**
 Nicholas E. Corrado, Josiah P. Hanna
 International Conference on Learning Representations (ICLR), 2024.
- [C7] **Deep Reinforcement Learning for Distribution Power System Cyber-Resilience via Distributed Energy Resource Control**
 Nicholas E. Corrado, Michael Livesay, Jay Johnson, Drew Levin
 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (IEEE Smart-GridComm), 2023.
- [C8] **Simulation-Acquired Latent Action Spaces for Dynamics Generalization**
 Nicholas E. Corrado, Yuxiao Qu, Josiah P. Hanna
 Conference on Lifelong Learning Agents (CoLLAs), 2022.

Workshop Papers.....

- [W1] **Distributionally Robust Multi-Task Reinforcement Learning via Adaptive Task Sampling**
 Nicholas E. Corrado, Wenyuan Huang, Josiah P. Hanna
 Automated RL Workshop @ RLC, 2026 **(Oral)**.
 🏆 **Best Paper Award**
- [W2] **Thinking is Another Form of Control**
 Josiah P. Hanna, Nicholas E. Corrado
 Finding the Frame Workshop at the Reinforcement Learning Conference (RLC), 2025 **(Oral)**.
 🏆 **“Most-Thought-Provoking Paper”**
- [W3] **On-Policy Policy Gradient Reinforcement Learning Without On-Policy Sampling**
 Nicholas E. Corrado, Josiah P. Hanna
 Finding the Frame Workshop at the Reinforcement Learning Conference (RLC), 2025.

HONORS & AWARDS

► Gold Reviewer @ ICML 2026 (Top 25%)	2026
► Top Reviewer Award @ NeurIPS 2024 (Top 8%)	2024
► Sandia Employee Recognition Award. Awarded to < 10% of the Sandia workforce	2023
► UW-Madison CS Department Scholarship (\$3000). Awarded to top graduate applicants.	2019
► John O. Blumberg Memorial Scholarship (\$1000). Awarded to the top math major.	2019
► Pennsylvania Space Grant Consortium Scholarship (third time, \$1500). Research funding.	2019
► Emil Sanielevici Scholarship (\$4000). Research funding.	2018
► Pennsylvania Space Grant Consortium Scholarship (second time, \$1500). Research funding.	2018
► J&M Bigos Memorial Scholarship (\$10,000). Awarded for academic excellence.	2018
► Sigma Pi Sigma Physics Honor Society	2018
► American Physical Society DPF Travel Award (\$200)	2017
► Peter F.M. Koehler Award (\$500). Awarded to the top physics major.	2017
► Brackenridge Summer Research Fellowship (\$3500). Research funding.	2017
► Rebecca Dytman Scholarship (\$10,000). Awarded for academic excellence in physics and astronomy.	2017
► Pennsylvania Space Grant Consortium Scholarship (first time, \$1500). Research funding.	2017

TALKS

- ▶ *On-Policy Policy Gradient Reinforcement Learning Without On-Policy Sampling* 2023
University of Edinburgh RL Reading Group [\[video\]](#)

ADVISING

- ▶ Harry Huang (Undergraduate, University of Wisconsin-Madison, WISCERS program) 2025-2026
- ▶ Nora Tseng (Undergraduate, University of Wisconsin-Madison, WISCERS program) 2024
Next: MS @ UC San Diego
- ▶ Yuxiao Qu (Undergraduate, University of Wisconsin-Madison) 2021-2023
Next: PhD @ Carnegie Mellon University

TEACHING EXPERIENCE

- University of Wisconsin–Madison.....
- ▶ Research Mentor Program (as part of the Delta Program) Fall 2023
 - ▶ Teaching Assistant for *Mathematical Foundations of Machine Learning (CS 761)* Fall 2021
 - ▶ Head Teaching Assistant for *Intro to Computer Systems (CS 354)* Fall 2021
 - ▶ Teaching Assistant for *Problem Solving for Engineers (CS 310)* Spring 2021
 - ▶ Teaching Assistant for *Discrete Mathematics (CS 240)* Fall 2020
- University of Pittsburgh.....
- ▶ Teaching Assistant for *Quantum Mechanics (PHYS 1370)* Fall 2018

SERVICE

- ▶ MadML Seminar Co-Organizer (University of Wisconsin-Madison) 2025
 - ▶ Reinforcement Learning Reading Group Coordinator (University of Wisconsin-Madison) 2025, 2026
 - ▶ Graduate Student Mentor for [Wisconsin Science and Computing Emerging Research Stars \(WISCERS\)](#) 2024, 2025, 2026
 - ▶ Graduate Student Mentor for Fall 2025 graduate cohort (University of Wisconsin-Madison) 2025
 - ▶ Invited Panelist, [Demystifying Graduate School](#) (University of Wisconsin-Madison) 2024, 2025, 2026
 - ▶ [Sandia Machine Learning and Deep Learning \(MLDL\) Workshop](#). Designed a new RL competition. 2022
- Reviewing.....
- ▶ Reviewer, NeurIPS 2023, 2024, 2025, 2026
 - ▶ Reviewer, ICML 2024, 2025, 2026
 - ▶ Reviewer, ICLR 2024, 2025, 2026
 - ▶ Senior/Holistic Reviewer, RLC (Reinforcement Learning Conference) 2024, 2025, 2026
 - ▶ Reviewer, TMLR 2025, 2026
 - ▶ Program Committee, AAAI 2024, 2025, 2026
 - ▶ Program Committee, AAAI Alignment Track 2026
 - ▶ Reviewer, RA-L (Robotics and Automation Letters) 2024
 - ▶ Reviewer, ICRA 2024
 - ▶ Reviewer, EWRL 2026

MEDIA

- ▶ [Training a dog and training a robot aren't so different](#) 2023

TECHNICAL SKILLS

Machine Learning & Data Science: Python ◦ PyTorch ◦ NumPy ◦ Pandas ◦ Matplotlib ◦ Jupyter ◦ Anaconda
Software Engineering: C++ ◦ C ◦ Git ◦ Bash